Computer-Assisted Hermeneutics of Philip K. Dick's Corpus: Constructing a Personal Knowledge Base with SpaCy and Obsidian for Literary Analysis

Audin, Yann
yann.audin@umontreal.ca
Université de Montréal, Canada

Objectives

This project seeks to explore computer-assisted hermeneutics in Philip K. Dick's science-fiction short stories, novels, and journals, as well as in his Gnostic influences. The project involves developing a Python library that bridges raw textual data and metadata with the Obsidian text editor to automatically generate a knowledge base using hyperlinks, indexes, knowledge graphs, context, information and timelines. More globally, the project aims to become a reusable model for computer-assisted hermeneutics, grounded in robust theoretical principles.

Philip K. Dick

Philip K. Dick (1928 - 1982) is widely recognised as one of the most influential and important science-fiction authors of the 20th century. Dick was celebrated by fans and critics alike, and received multiple awards including a Hugo (1962, The Man in the High Castle) ; his work is still being adapted to the cinema forty years after his death. Despite literary successes and ongoing popularity, he faced constant financial difficulties and wrote prolifically to support himself and his relatives. He used (and abused) amphetamines to sustain the rhythm at which he wrote to make up for low rate of pay from science-fiction magazines. Two years after he attempted suicide while travelling in Canada, he had a religious experience in 1974 that informed his writing for the last eight years of his life. After meeting a woman wearing the Christian ichthys symbol, he started to hallucinate a pink beam of light and visions of early Christians. The effects lasted for weeks and Dick documented his experience and thoughts in journals he called "2-3-74" (referring to February and March 1974), excerpts of which were later published in The Exegesis of Philip K. Dick (edited by Pamela Jackson and Jonathan Lethem). Even before the event, Dick was well versed in Gnostic theology and early Church history.

Objectives

Phases of the Project

The project is structured onto three phases : the first (and current) phase centres around Dick's short stories and The Exegesis : we developed a metadata schema, tested various libraries for Named Entity Recognition, and evaluated tools for text and data visualisation. In the second part of the project, Philip K. Dick's novels and letters will be added to the corpus and analysed. A third and final phase will see the inclusion of Gnostic gospels and texts. The approach should be theorised and generalised for application to other literary corpora. Each stage of the process includes a hybrid analysis combining close and distant reading.

The Philip K. Dick Corpus

Philip K. Dick's bibliography includes 35 published novels, 10 posthumously released novels and 121 short stories, although the exact count varies because of classification debates. His letters were collected in 6 volumes and parts of his journal were published in 2011. The project is currently testing hermeneutic and computational methods on Dick's short stories and the Exegesis. We have gathered 115 of his short stories, comprising a total of 769 816 words (or 4 382 718 characters) – an average of 6 694,05 words per text.

Research Questions

This project is informed by two aspects of Philip K. Dick's writing : he named some of his characters after people he knew or even himself (for instance, in Mr Spaceship, What the Dead Men Say and A Terran Odyssey), and included important concepts and stories from Gnostic texts in his writing. Hence, this project aims to answer two questions of literary and historical interest :

1. How do recurring names in Philip K. Dick's science-fiction and nonfiction works reflect connec-

tions between his personal life, own mythology and literary output ?

1. How does Gnostic theology interact with Dick's science-fiction and journals ?

We investigate Philip K. Dick's work by cross-referencing his journals, letters and biographical data with his literary texts using automatically generated indexes, timelines, knowledge graphs, hyperlinks and networks. A particular point of interest is his self-constructed mythology, and the effect of the religious experience described in "2-3-74" on his life and work.

Methodology

The project follows an iterative process applied to each phase of the corpus development :

Corpus preparation

The texts are saved as .txt files and the metadata are structured in a json file using multiple sources for accuracy.

Development of a Python-based software

The library interfaces between raw textual data and *Obsidian*. The project aims at being reusable and extendable beyond the suggested corpus.

Database generation

Using a dictionary of Gnostic terms and Named Entity Recognition with SpaCy, a database of words of interest and their specific occurrences in the corpus is created.

Conversion to Markdown for Obsidian

The cleaned texts are augmented with markdown hyperlinks that lead to virtual index cards of names and Gnostic concepts. Timelines and networks are automatically generated in an *Obsidian*-readable format.

Mixed-method analysis

The enhanced corpus is studied through assisted close reading and distant reading within the literary knowledge base using *Obsidian*.

Implication

This project takes full advantage of ready-made and custom-made tools to create a framework for computational literary studies. It preserves the humanistic values of the hermeneutic process, while leveraging digital media and algorithmic tools. More specifically, the method showcases how classical Natural Language Processing can be used to meaningfully augment literary corpora which are usually considered too large for traditional literary analysis. It serves as a model to combine close and distant reading to better understand the interplay of an author's literary production, influences and personal mythology.
This presentation will introduce Philip K. Dick's corpus and outline our methodology from a theoretical and technical perspective. We will present the knowledge graphs and knowledge base, as well as the results for the first phase of the project

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